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Fall 2016 - 2017

Q1:

(a) $\left(\frac{1+e^x}{e^x}\right)\frac{dy}{dx} = y$

$\int \frac{1}{y}\,dy = \int \frac{e^x}{1+e^x}\,dx \rightarrow \ln|y| = \ln|1+e^x| + C.$

(b) $\frac{\partial M}{\partial y} = -\sin y \qquad \frac{\partial N}{\partial x} = -\sin y \qquad \therefore$ Exact

$\int M\,dx + \int N\,dy = C$

$\frac{1}{2}x^2 + x\cos y + \ln|\sec y| - x\cos y = C$

$\frac{1}{2}x^2 + \ln|\sec y| = C.$

(c) xx

(d) $\mu = e^{\int \frac{-x-1}{2x}(-2)} = e^{\int 1 + \frac{1}{x}} = e^{x+\ln x} = e^x \cdot e^{\ln x} = xe^x$

$y^{-2} = \frac{1}{xe^x}\left[\int xe^x\left(\frac{-3}{x}\right)(-2)\,dx + C\right]$

$= \frac{1}{xe^x}\left[6e^x + C\right] = 6x^{-1} + \frac{C}{xe^x}.$

Q2:

(a) $m^3 - 2m^2 - 5m + 6 = 0$

$m_1 = -2 \quad m_2 = 3 \quad m_3 = 1$

$y = C_1e^{-2x} + C_2e^{3x} + C_3e^x.$

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(b) $m^2 - m - 2 = 0$

$(m+1)(m-2)$

$m_1 = -1 \quad m_2 = 2$

$y_{c.f} = C_1 e^{-x} + C_2 e^{2x}$

$y_{p.I} = Ax + B + Ce^{-x}$

$y' = A - Ce^{-x}$

$y'' = Ce^{-x}$

$\therefore Ce^{-x} - A + Ce^{-x} - 2Ax - 2B - 2Ce^{-x} = 6x + 6e^{-x}$

$-2C = 6 \rightarrow C = -3 \qquad -2A = 6 \rightarrow A = -3$

$-A - 2B = 0 \rightarrow -A = 2B \rightarrow B = \frac{-A}{2} = \frac{3}{2}$

$y_{p.I} = -3x + \frac{3}{2} - 3e^{-x}$

$y_{g.s} = y_{c.f} + y_{p.I} = C_1 e^{-x} + C_2 e^{2x} - 3x + \frac{3}{2} - 3e^{-x}$

(c) $m^2 + 1 = 0$

$m_{1,2} = \pm i$

$y_{c.f} = e^{0x}[C_1 \cos x + C_2 \sin x]$

$u = \cos x \quad v = \sin x$

$u' = -\sin x \quad v' = \cos x$

$uv' - vu' = 1$

$AX(u) = \int \frac{-vf(x)}{1} = \int -\sin x \cdot \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} = -\int \frac{\sin^2 x}{\cos^2 x}$

$= -\int \frac{1 - \cos^2 x}{\cos^2 x} = -\int \frac{1}{\cos^2 x} - 1 = -(\tan x - x) + C$

$BX(v) = \int \frac{uf(x)}{1} = \int \cos x \cdot \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} = \int \tan x = \ln|\sec x| + k.$

$y_{g.s} = e^{0x}[C_1 \cos x + C_2 \sin x] - \tan x + C + C + \ln|\sec x| + k.$

(d) $y'' + \frac{1}{x} y' - \frac{1}{x^2} y = \frac{1}{x}$

$y = Vx$

$\frac{dy}{dx} = \frac{dv}{dx} x + V$

$\frac{d^2y}{dx^2} = \frac{d^2v}{dx^2} x + \frac{dv}{dx} + \frac{dv}{dx}$

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$$\frac{d^2v}{dx^2}x + 2\frac{dv}{dx} + \frac{dv}{dx} + \frac{v}{x} - \frac{v}{x} = \frac{1}{x}$$

$$x\frac{d^2v}{dx^2} + 3\frac{dv}{dx} = \frac{1}{x} \qquad \frac{dv}{dx} = z \qquad \frac{d^2v}{dx^2} = \frac{dz}{dx}$$

$$x\frac{dz}{dx} + 3z = \frac{1}{x}$$

$$\frac{dz}{dx} + \frac{3}{x}z = \frac{1}{x^2}$$

$$\mu = e^{\int \frac{3}{x}} = e^{3\ln x} = e^{\ln x^3} = x^3$$

$$z = \frac{1}{x^3}\left[\int x^3 \cdot \frac{1}{x^2}dx + c\right] = \frac{1}{x^3}\left[\frac{1}{2}x^2 + c\right]$$

$$\frac{dv}{dx} = \frac{1}{2x} + \frac{c}{x^3} \rightarrow \int dv = \int \frac{1}{2x} + \frac{1}{x^3}c\, dx$$

$$v = \frac{1}{2}\ln|x| + \left(\frac{-1}{2}x^{-2}c\right) + k$$

$$\frac{y}{x} = \frac{1}{2}\ln|x| + \left(\frac{-1}{2}x^{-2}c\right) + k$$

$$\# \left\{ y = \frac{x}{2}\ln|x| - \frac{1}{2}xc + kx \right\} \#$$

Q3:7

(a) $\int_0^\infty e^{-3t}\left(\frac{1}{2} - \frac{1}{2}\cos 2t\right)dt$

$$\frac{1}{s}\left[\frac{1}{2(s+3)} - \frac{1}{2}\frac{(s+3)}{(s+3)^2+4}\right].$$

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(b)

$$s^2y(s) - s\,y(0) - y'(0) - 6\,sy(s) - 6\,y(0) + 8\,y(s) = \frac{1}{s-3}$$

($y(0)$ → Zero, $y'(0)$ → 2, $y(0)$ → Zero)

$$y(s)\left[s^2 - 6s + 8\right] = \frac{1}{s-3} + \frac{2}{1} = \frac{1+2s-6}{s-3}$$

$$y(s) = \frac{2s-5}{(s-3)(s-4)(s-2)} = \frac{A}{s-3} + \frac{B}{s-4} + \frac{C}{s-2}$$

$$\boxed{A = -1} \quad \boxed{B = \frac{3}{2}} \quad \boxed{C = \frac{-1}{2}}$$

$$y(s) = -\frac{1}{s-3} + \frac{3}{2}\frac{1}{s-4} - \frac{1}{2}\frac{1}{s-2}$$

$$y(t) = -e^{3t} + \frac{3}{2}e^{4t} - \frac{1}{2}e^{2t}$$

(c) (i) $f(t) = \int_0^t \tau \sin\tau \, d\tau$

$$f(s) = \frac{1}{s}\left[\frac{-d}{ds}\frac{1}{s^2+1}\right] = \frac{1}{s}\left[\frac{2s}{(s^2+1)^2}\right]$$

(ii) $f(t) = \sin 5t \cos 3t + \cosh 3t$

$$f(t) = \frac{1}{2}[\sin 8t - \sin 2t] + \cosh 3t$$

$$f(s) = \frac{1}{2}\left[\frac{8}{s^2+64} - \frac{2}{s^2+4}\right] + \frac{s}{s^2-9}$$

(iii) $f(t) = u_4(t)\, e^t$

$$f(s) = e^{-4s}\frac{1}{s-1}$$

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Q_4: (i) $F(s) = \ln(s^2+16) - \ln(s^2-5s+6)$

$$F'(s) = \frac{2s}{s^2+16} - \frac{2s-5}{s^2-5s+6}$$

$$= 2\left[\frac{s}{s^2+16}\right] - 2\left[\frac{s}{s^2-5s+6} - \frac{5}{s^2-5s+6}\right]$$

$$= 2\left[\frac{s}{s^2+16}\right] - 2\left[\frac{s}{s^2-5s+\frac{25}{4}+6-\frac{25}{4}} - \frac{5}{s^2-5s+\frac{25}{4}+6-\frac{25}{4}}\right]$$

$$= 2\left[\frac{s}{s^2+16}\right] - 2\left[\frac{s}{(s+2.5)^2-\frac{1}{4}} - \frac{5}{(s+2.5)^2-\frac{1}{4}}\right]$$

$$= 2\cos 4t - 2\left[e^{2.5t}\cosh(\tfrac{1}{2})t - \tfrac{5}{2}e^{2.5t}\sinh(\tfrac{1}{2})t\right].$$

$\downarrow$

$$-t\,f(t) =$$

$$f(t) = \frac{2\cos 4t - 2\left[e^{2.5t}\cosh(\frac{1}{2})t - \frac{5}{2}e^{2.5t}\sinh(\frac{1}{2})t\right]}{-t}.$$

(ii) $$f(s) = \frac{s}{s^2+4s+8} = \frac{s}{s^2+4s+8+4-4} = \frac{s}{(s+2)^2+4}$$

$$f(t) = e^{-2t}\cosh 2t$$

(iii) $$F(s) = e^{-2s}\left[\frac{1}{s^2-10s+16}\right] = e^{-2s}\left[\frac{1}{(s^2-5)^2-9}\right]$$

$$= e^{-2s}\left[\tfrac{1}{3}e^{5t}\sinh 3t\right] =$$

$$f(t) = \tfrac{1}{3}u_2(t)\; e^{5(t-2)}\sinh 3(t-2).$$

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[b] $0<x<2$ Sine

↓ odd → b_n? a_0, a_n = Zero

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{\tau} x$$

$$b_n = \frac{1}{2}\int_0^2 x^2 \sin\tfrac{1}{2}n\pi x$$

x^2	$\sin\frac{1}{2}n\pi x$
$2x$	$\frac{\cos\frac{1}{2}n\pi x}{n}$
2	$\frac{-\sin\frac{1}{2}n\pi x}{n^2}$
0	$\frac{-\cos\frac{1}{2}n\pi x}{n^3}$

$$= \frac{1}{2}\left[\frac{x^2\cos\frac{1}{2}n\pi x}{n} + \frac{2x\sin\frac{1}{2}n\pi x}{n^2} - \frac{\cos\frac{1}{2}n\pi x}{n^3}\right]_0^2$$

$$= \frac{1}{2}\left[\left(\frac{4\cos\frac{n\pi}{2}}{n} + \frac{4\sin n\pi}{n^2} - \frac{\cos n\pi}{n^3}\right)\right]$$

($\frac{4\sin n\pi}{n^2}$ → Zero)

$$= \frac{1}{2}\left[\frac{4(-1)^n}{n} - \frac{(-1)^n}{n^3}\right]$$

$$f(x) = \sum_{n=1}^{\infty}\left(\frac{1}{2}\left[\frac{4(-1)^n}{n} - \frac{(-1)^n}{n^3}\right]\right)\sin\frac{n\pi x}{2}$$

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